Implicit Neural Representation of Graphons for Extremal Graph Theory

(/pdf?id=JUhuwRQlDV)

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Verify Author List: I have double-checked the author list and understand that additions and removals will not be allowed after the abstract submission deadline.

TL;DR: We use implicit neural representation on graphon to solve the extremal graph theory problems.

Abstract:
Machine learning is opening new directions in mathematical discovery, from automatic provers for Olympiad problems to progress on long-standing conjectures. We propose a neural approach to asymptotic extremal problems in graph theory, which study the limiting behaviour of graph optimisation as the number of vertices tends to infinity. Such problems can often be reformulated as continuous optimisation over *graphons*, symmetric measurable functions $[0, 1]^2 \to [0, 1]$. We represent graphons as implicit neural networks and optimise the resulting objectives by gradient descent. Because optimal graphons are typically discontinuous (often step-like), standard neural architectures perform poorly; to address this, we use a diffusion-inspired design with multi-scale sinusoidal input encodings. Also, to enforce the constraint (e.g., fixed edge density) in such a graphon-optimisation problem, we incorporate a solver for an empirical version of the constraint into the architecture, and backpropagate through the solver via implicit differentiation. For some generalised Turán problems and domination exponent problems that previously required extensive human effort, our method rediscovers known optimal graphons without human intervention. For some open problems of the same kind, it finds novel candidate optima that may be of interest to the mathematics community.

Primary Area: Applications->Everything Else

Keywords: Implicit Neural Representation, Extremal Graph Theory, AI for Math

Ethics Agreement: I certify that all co-authors of this work have read and are committed to adhering to the Call for Papers, Author Instructions, Research Ethics, and Peer-review Ethics.

LLM Policy: This submission requires Policy A.

Proceedings-only Option: If this paper is accepted, the authors tentatively plan to present it in person at the conference (as a poster and, if selected, as an oral).

Ready For LLM Feedback: The submitted PDF is ready for LLM feedback.

LLM Feedback Request: Taeyoung Kim (/profile?id=~Taeyoung_Kim2)

CORRECTION REQUEST: Reason: We would like to claim an exemption, because none of the authors are qualified according to the definition given in the Peer Review FAQ.

CORRECTION REQUEST: Explanation: Taeyoung Kim and Joonkyoung Lee has one paper on ICML and IJCAI, respectively. Jineon Baek has no publication on AI conferences. Hongseok Yang has multiple publications on AI conference, but none of those as primary author.

Reciprocal Reviewing Status: This submission is NOT exempt from the Reciprocal Reviewing requirement. (We expect most submissions to fall in this category.)

Reciprocal Reviewing Author: Taeyoung Kim (/profile?id=~Taeyoung_Kim2)

Submission Number: 10609

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Paper Decision

Decision by Program Chairs 01 May 2026, 00:57 (modified: 01 May 2026, 03:07)
Program Chairs, Authors Revisions (/revisions?id=Cyk5NrTF1u)

Decision: Reject

Comment:

This paper presents a neural approach to study the limiting behaviour of graph optimization as the number of vertices tends to infinity by reformulating it as continuous optimization over graphons and representing graphons as implicit neural networks. The diffusion-inspired design with multi-scale sinusoidal input encodings and the solver for an empirical version of the constraint are proposed. Three of the four reviewers provide weakly positive ratings by considering the mathematical soundness, good presentation, and significance toward approaching highly abstract mathematical problems through optimization. One reviewer remains unconvinced that this work will have any meaningful impact on extremal graph theory, and a lack of comparison in the abstract and introduction to recent AI4Math advances.

Official Review of Submission10609 by Reviewer By6w

Official Review by Reviewer By6w 12 Mar 2026, 23:21 (modified: 08 Apr 2026, 23:40)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer By6w
Revisions (/revisions?id=8ivZlRak97)

Summary:

Certain optimization problems admit graphs as their solutions. Because graphs are combinatorial objects, direct comparison between candidate solutions is often cumbersome. A standard way to circumvent this difficulty is to formulate the problem in terms of homomorphism densities, which quantify the number of structure-preserving maps from one graph into another. This representation makes it possible to express in a compact and mathematically tractable manner questions concerning the existence of graphs satisfying prescribed constraints, as well as questions comparing different graph families.

Since these quantities depend on the size of the underlying graph, it is natural to ask how such optimization problems behave in the limit of large numbers of vertices. In this asymptotic regime, the corresponding limit objects are graphons. Graphons, can be understood as analytic extensions of finite graphs that arise when the number of vertices tends to infinity.

In this manuscript, the authors propose a method for approximating graphon solutions to extremal graph problems. Their approach relies on implicit representation networks as a parametrization of candidate graphons. The manuscript also addresses some of the main technical difficulties inherent to this framework, in particular those related to the enforcement of constraints and the estimation of homomorphism densities. Finally, the proposed methodology is evaluated on both solved and open nontrivial instances, which allows for a comparison with known results as well as an assessment of its performance in less understood settings.

Strengths And Weaknesses:

Regarding Soundness:

The manuscript is mathematically sound and the notation is, for the most part, clear and consistent. The formulation of graph optimization problems in terms of homomorphism densities, together with their extremal counterparts, is presented in a simple and accessible way. The optimization framework based on neural networks is also introduced in a reader-friendly manner. In addition, the reduction in the computational cost of homomorphism density estimation becomes a particularly useful aspect of the proposed approach.

Regarding Presentation:

I found the presentation to be overall good. In particular, the manuscript succeeds in motivating the passage from discrete graph problems to their extremal counterparts, and this is presented in a clear and accessible way despite the abstract nature of graphons. The inclusion of graphon visualizations also strengthens the exposition, as these figures are well executed and help give coherence to the manuscript.

That said, the presentation could benefit from a more explicit discussion of why homomorphism densities provide the appropriate language for the formulation of graph problems. Although the framework is introduced clearly, further motivation in this direction would help emphasize its role and importance.

A more detailed explanation of graphons themselves could also improve the manuscript for non-expert readers. While most readers will already be familiar with finite graphs, it would be helpful to explain more carefully how graphons arise naturally as limit objects when the number of vertices tends to infinity. Since graphons are the central objects of the manuscript, additional intuition at this stage would likely make the later developments easier to follow.

A similar comment applies to the implicit representation of graphons. The use of sinusoidal residual components in the approximation architecture would benefit from a stronger motivation within the narrative. Beyond the interpretation in terms of low- and high-frequency behavior, the exposition would be strengthened by a more concrete justification for this architectural choice, particularly in light of existing work in this direction (see https://arxiv.org/pdf/2211.03329 (https://arxiv.org/pdf/2211.03329)).

Regarding Significance:

The idea of approaching highly abstract mathematical problems through optimization is, from my point of view, particularly compelling. In this context, the use of neural networks to approximate abstract limit objects such as graphons may become a valuable tool, especially when one is interested in exploring or providing insights toward difficult mathematical conjectures.

Regarding Originality:

While the proposed graphon network is not entirely original in itself, its combination with homomorphism density estimation and constrained optimization updates, together with its application to abstract problems such as extremal graph theory, results in a methodology that feels both original in scope and modern in perspective.

Soundness: 4: excellent
Presentation: 3: good
Significance: 3: good
Originality: 3: good
Key Questions For Authors:
I am concerned about results on table 8.

-Which kind of normalization was applied to the density in table 8?

-I have a question regarding table 8 caused by the exponent of order e-25, which is extremely low even for a conservative difference. Therefore, I'm not sure this result is numerically meaningful.

I understand no problem solving is claimed and the result is just presented as a possible counterexample for the conjecture. However, could you clarify to me why these numbers are still interpretable? It looks like the counterexample proposed in Fig 8, is sparse in some sense and could benefit from some normalization (See https://arxiv.org/pdf/1601.07134 (https://arxiv.org/pdf/1601.07134)). Maybe the homomorphism density estimation works with small motifs (e.g. K2, K3) but makes convergence fail with larger ones (e.g. Desargues and Heawood).

Limitations:
The method seems to work well with problems depending on small motifs. If the either objective function or constraints involve larger graphs, I am uncertain about the results here presented.

Overall Recommendation: 4: Weak accept: Technically solid paper that advances at least one sub-area of AI, with a contribution that others are likely to build on, but with some weaknesses that limit its impact (e.g., limited evaluation). Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Compliance With LLM Reviewing Policy: Affirmed.
Code Of Conduct Acknowledgement: Affirmed.
Final Justification:
The manuscript presents a novel approach to solving difficult and highly abstract mathematical problems whose solutions are graphons.

The authors are going out on a limb here, and in that context it is not surprising that not every experiment yields a fully conclusive outcome. If one considers the manuscript as a whole, however, one finds a difficult problem presented in an understandable way, an innovative solution, and numerous detailed experiments supporting its usefulness. I therefore regard the uncertainty in the final experiment as a reasonable limitation.

Regarding the presentation, I agree that including Table 8 and Figure 8 in the main body is not entirely appropriate and that it would be better placed in an appendix. Despite the interest of proposing a counterexample to the conjecture using a graph as relevant as the Desargues graph, the authors are ultimately unable to make a concrete claim about the proposed solution. In that sense, this part goes somewhat against the cohesion of the manuscript.

For this reason, I would revise my score to 4. The limitation identified above does not invalidate the rest of the manuscript, whose results remain very interesting.

Rebuttal by Authors

Rebuttal
by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))
31 Mar 2026, 15:35 (modified: 31 Mar 2026, 22:24)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors
Revisions (/revisions?id=HcoVu1yGGN)

Rebuttal:
We thank the reviewer for the positive assessment and helpful comments.

1. That said, the presentation could benefit from a more explicit discussion of why homomorphism densities provide the appropriate language for the formulation of graph problems. (...) Since graphons are the central objects of the manuscript, additional intuition at this stage would likely make the later developments easier to follow.

Homomorphism densities are a natural language for extremal graph problems because they are size-normalised statistics of fixed patterns. They let one express objectives and constraints in a way that is comparable across graphs of different sizes, and they play a central role in dense graph limit theory, quasirandomness, and regularity-based approximation. In applied settings, closely related motif densities are also used in network analysis to summarise local organisation and compare large social, biological, and communication networks through the frequencies of small patterns such as edges, triangles, and other motifs.

In the revised version of the paper, we will add a short discussion along these lines. We will also explain why graphons arise naturally as limiting objects when the homomorphism densities of every fixed graph converge along a sequence of dense graphs, and we will include intuitive examples showing how familiar graph structure is reflected in a graphon.

2. Which kind of normalization was applied to the density in table 8?

Among inequalities of the form $t(H_1, W)^{r_1} \geq t(H_2, W)^{r_2}$, we call such an inequality a vertex-normalised density inequality if r_1 and r_2 are computed from $v(H_1)$ and $v(H_2)$, and an edge-normalised density inequality if they are computed from $e(H_1)$ and $e(H_2)$. In Table 8, the inequality we are interested in is $t(H, W) \geq t(D, W)^{v(H)/v(D)}$, a vertex-normalised inequality, and the normalised difference refers to $t(H, W) - t(D, W)^{v(H)/v(D)}$. Similarly, the normalised homomorphism density in Table 6 corresponds to the edge-normalised homomorphism density, since $\binom{s}{2}$ and $\binom{r}{2}$ are the numbers of edges in K_s and K_r. We will update the description to make this normalisation explicit in the revised version.

3. I have a question regarding table 8 caused by the exponent of order e-25, which is extremely low even for a conservative difference. Therefore, I'm not sure this result is numerically meaningful. (...) However, could you clarify to me why these numbers are still interpretable?

See our response to Q2 of Reviewer yUB7.

4. It looks like the counterexample proposed in Fig 8, is sparse in some sense and could benefit from some normalization (See https://arxiv.org/pdf/1601.07134 (https://arxiv.org/pdf/1601.07134)).

Thank you for the reference. We note that, although the graphon in Fig. 8 has relatively small edge density, it still lies in the dense-graph regime considered in our paper. A graph sequence is usually called sparse when its number of edges grows subquadratically, i.e. $e(G_n) = o(v(G_n)^2)$. Under the dense graphon formalism used here, such sequences collapse to the zero graphon in the limit, so obtaining an informative limit object requires a different sparse-limit framework. Several such frameworks have been proposed, including graphon-process and related approaches (e.g. https://arxiv.org/abs/1205.4356 (https://arxiv.org/abs/1205.4356), https://arxiv.org/abs/1504.00858 (https://arxiv.org/abs/1504.00858)). We agree that connecting these sparse-limit formalisms with implicit neural representations would be an interesting direction for future work.

5. Maybe the homomorphism density estimation works with small motifs (e.g. K2, K3) but makes convergence fail with larger ones (e.g. Desargues and Heawood).

It is true that larger motifs make homomorphism-density estimation harder, but the effect is not catastrophic. In our implementation, for a graph with $v(H)$ vertices, the computational cost scales like $v(H)^2$ neural-network forward passes, so under the same memory budget the batch size must decrease roughly quadratically as the graph size grows. To compensate for this, we used gradient accumulation, as described in Section C of the supplementary material. At first sight, one may expect a curse of dimensionality because the integral is over $[0, 1]^{v(H)}$. However, since the integrand is 1-dimensional and bounded in $[0, 1]$ and we are estimating this integrand, the Monte Carlo convergence rate in absolute error remains $O(1/\sqrt{N})$ for the number N of samples, and this rate is independent of the dimension of $[0, 1]^{v(H)}$.

In the Desargues-Heawood experiment, the main difficulty is relative rather than absolute accuracy: if the true mean is q, then for a bounded integrand in $[0, 1]$ the relative Monte Carlo error is on the order of $\sqrt{(1-q)/(qN)}$, so achieving the same relative accuracy comparable to the $q = \frac{1}{2}$ case for a general q requires on the order of q^{-1} more samples.

Replying to Rebuttal by Authors

Rebuttal Acknowledgement by Reviewer By6w

Rebuttal Acknowledgement by Reviewer By6w 04 Apr 2026, 04:24

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (b) Partially resolved - I have follow-up questions for the authors.

Reasons:

Thank you for taking the time to reply. From my perspective, the value of the work does not rely on the particular example in Table 8. I find value in an approximate tool for solving complex mathematical problems.

However, I still think the non-representativeness of the Table 8 results needs to be addressed. I am still not sure what could be causing those numbers to reach such magnitudes. I also see this as a great opportunity to improve the work: if it is a numerical issue, how can it be addressed?

I understand that the homomorphism density between H and W is already very low and does not depend on the chosen graph G. However, you mentioned that the conjecture is formulated for 3-regular graphs with specific girth. I do not know the reasons for choosing the Desargues graph, but since this is a computational experiment, nothing prevents you from trying a wider range of graphs; perhaps that would yield more convincing results.

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Replying to Rebuttal Acknowledgement by Reviewer By6w

Reply Rebuttal Comment by Authors

Reply Rebuttal Comment

by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))

07 Apr 2026, 21:25 (modified: 07 Apr 2026, 22:05)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=OeZPwZD3yI)

Comment:

Regarding H.-D., we will tone down our claim, avoiding the words "evidence" or "counterexample", and will simply say that the found graphon is a candidate optimal solution of $\min_W t(H,W) - t(D,W)^{v(H)/v(D)}$. We think that understanding the structure of an optimising graphon here is valuable because of the importance of the Desargues graph in this context. The Desargues graph satisfies a weakly norming property, i.e., the homomorphism density function $t(H,\cdots)$ is convex on the set of graphons (https://arxiv.org/abs/1611.05784 (https://arxiv.org/abs/1611.05784)). Weakly-norming graphs are known to dominate all of their subgraphs F in the sense that $t(H,W)^{1/e(H)} \geq t(F,W)^{1/e(F)}$. So, weakly norming graphs are positioned close to the top of domination relations and are strong candidates for disproving the conjectured domination inequality. Moreover, if the conjectured domination inequality holds for the Desargues graph, it also implies that the Heawood graph is Sidorenko, $t(H,W) \geq t(K_2,W)^{e(H)}$, concluding one of the hardest instances in the long-lasting conjecture in extremal graph theory. A 3-regular weakly norming graph with girth at least 6 is extremely rare; perhaps very few exist and the Desargues graph seems to be the only one that is small enough to be handled.

Before finishing, we re-emphasise that we made the following new experimental findings in extremal graph theory using our approach described in pages 7 and 8 of the paper.

- Tab 3, Fig 3: On the problem of minimising the C_5 density under a fixed edge density p, our approach found candidate optimal graphons with unexpected banded structure for $p \neq 1-1/k$. See Fig 3(d)-(f). Understanding the structure of an optimal graphon is an important topic in extremal graph theory. For the $p = 1-1/k$ case, the approach found experimental support for the conjectured optimal C_5 density by Bennett et al. (2020) (https://arxiv.org/abs/1803.00165 (https://arxiv.org/abs/1803.00165)), and provided experimental evidence that an optimal graphon in that case is likely to have the k-partite structure. See Fig 3(a)-(c).
- Tab 4, Fig 4: Using our approach, we analysed the C_7-density minimisation problem under a fixed edge-density constraint for the first time to our knowledge. We generalised the conjecture of Bennett et al. on the C_5 density (described in Appendix D). Using our approach, we found experimental support for the conjecture. Also, candidate optimal graphons for edge densities $p = 1-1/k$ found by our approach have the k-partite structure (Fig 4(a)-(c)), but those for $p \neq 1-1/k$ have non-trivial structure, such as banded structure (Fig 4(d)-(f)).
- Tab 5, Fig 5: Our study of the curve $p \longmapsto \min_{W:t(K_2,W)=p} t(H,W)$ for a range of graphs H provides a new experimental observation about the shape of this curve; when H is close to being a complete graph, the curve tends to become concave and this concavity fails as H gets further from being complete. This suggests an interesting new question about mathematically characterising the graph H that makes the curve concave. Also, our experiments for $H = H_6$ in Fig. 5(a) show that optimal graphons for H_6 and $p = 1-1/k$ are not achieved by k-partite graphons or constant graphons.
- Tab 9, Fig 9, Appendix E: As far as we know, the minimisation of the Petersen density under an edge-density constraint has not been studied before. Our approach experimentally provides information about what optimal graphons might look like and where the optimal values lie.

- Large Deviation Application: During the rebuttal period, we applied our approach to the following graphon-optimisation problem, which is studied by (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) in order to analyse conditioned random graph models:

$$\inf_W \int \left(W(x,y) \log \frac{W(x,y)}{p} + (1 - W(x,y)) \frac{1 - W(x,y)}{1-p} \right) dx\, dy \,\text{s.t.}\; t(K_3, W) \geq r^3.$$

By Chatterjee and Varadhan's large deviation theory, the optimising graphons characterise the probability distribution of the Erdős–Rényi random graph $G(n,p)$ conditioned on the triangle density being at least r^3. We applied our approach to $(p,r) \in \{0.05, 0.1\} \times \{0.4, 0.5, 0.6\}$, and consistently found a two-block structured graphon (also known as stochastic block model) as a candidate optimal solution, and such a two-block graphon outperformed the example in (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) experimentally. Our finding gives rise to a conjecture that a two-block graphon is an optimiser. Details of our experiments are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba (https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba)

Official Review of Submission10609 by Reviewer uEHG

Official Review by Reviewer uEHG 12 Mar 2026, 07:06 (modified: 07 Apr 2026, 12:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uEHG

Revisions (/revisions?id=YxavqJCl7m)

Summary:

This work proposes a method for solving asymptotic extremal graph theory problems by optimizing over graphons. The authors implicitly represent graphons as neural networks and minimize objectives that are functions of homomorphism densities, potentially under constraints (i.e., edge density). The paper focuses on two special cases, namely minimizing a homomorphism density under fixed edge density, and minimizing normalized differences of homomorphism densities. To this end, the authors introduce a specifically crafted residual MLP structure based on sinusoidal encodings of the input on increasing frequency/scale over the layers, as well as a differentiable solver for the empirical edge density constraint. Empirically, the method rediscovers several optimal graphons for triangle, clique, and 5-cycle minimization under edge density constraints. Further, it generates plausible approximations to candidate optima for unresolved problems, including 7-cycle minimization and a conjectured inequality involving the Heawood and Desargues graphs.

Strengths And Weaknesses:

Strengths:

- The paper studies an unusual and potentially interesting application of machine learning to extremal graph theory problems, and the idea of implicitly representing graphons as neural networks is conceptually clear and nontrivial.
- The method is capable of recovering several known or conjectured extremal graphons, which gives this search heuristic some credibility.

Weaknesses:

- To me, the presented method seems highly specialized to the exact problem class studied. Namely, the constraint is handled through a scalar shift in the last layer, and it is unclear how this would generalize to constraints other than edge density. The framing of the architecture as being "diffusion-inspired" (l. 25 abstract, l. 127-130 on the right) feels overstated, the architecture is more accurately described as a hand-crafted multiscale sinusoidal neural network. The exact inductive bias looks quite strongly feature-engineered for this domain, potentially even for the limited number of examples studied in the experimental section.
- The paper does not make a convincing case that a neural representation is preferable to simpler parameterizations such as SBMs, wavelets, or maybe even heuristics like decision trees (also considering that graphons are only functions on a 2D domain, so function fitting would not suffer from the curse of dimensionality). The empirical evaluation does not seem to compare against non-neural baselines for graphon search. This makes it hard to judge for me if the proposed method should be preferred or is merely one workable among many parameterizations.

- Re the open problem about the Heawood graph conjecture (l. 408-414 right, fig. and table 8), the evidence for an actual counterexample seems quite weak. Both densities are in the range of 10^{-24}—10^{-25}, and the estimation confidence intervals overlap strongly (in fact, even one contains the other). Hence, it would seem plausible to me that the observed differences are merely estimation uncertainty. Furthermore, reporting the equivalent normalized quantities in table 8 (i.e. $t(H,W)^{1/v(H)} - t(D,W)^{1/v(D)}$) might make the magnitude of the effect easier to interpret.
- This is rather superficial in comparison, but the exposition reads as somewhat unfocused. Lemma 3.1 appears overemphasized compared to its actual role (the work does not seem to use this explicit formula anywhere else, and an autodiff framework can calculate this as well), or mentions of colormap/font conventions in the experimental results section of the main body (l. 260 onwards, right).

Soundness: 2: fair
Presentation: 2: fair
Significance: 2: fair
Originality: 2: fair
Key Questions For Authors:

- Could you refer to the points in "Weaknesses"?
- The overall contribution is not entirely clear to me. On the one hand, I do not yet see this as a framework that would transfer beyond the fairly specific setting of this paper, which makes its hard for me to assess relevance to a broader ML audience. On the other hand, I am also unsure what the precise value proposition would be for a mathematician: I am by no means an expert in extremal graph theory, but I would expect certifiable or more symbolic/combinatorial approaches (say, tools like flagmatic https://lidicky.name/flagmatic/ (https://lidicky.name/flagmatic/)) to be more helpful as they can provide verifiable bounds rather than rough numerical approximations. Could you maybe comment more explicitly on where you see this method relative to existing tools, and in what sense you might expect this to assist progress in mathematics? I would be happy to update my score if the authors clarify this point convincingly.

Limitations:
yes

Overall Recommendation: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility, incompletely addressed ethical considerations, or writing so poor that it is not possible to understand its key claims.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Compliance With LLM Reviewing Policy: Affirmed.
Code Of Conduct Acknowledgement: Affirmed.
Final Justification:
While the rebuttal addressed some concerns regarding the positioning of the work relative to existing approaches, I remain unconvinced that this work will have any meaningful impact on extremal graph theory. In addition, the framing of the paper, particularly the comparison in the abstract and introduction to recent AI4Math advances, seems quite weak, as that the core contribution is parametrically fitting a graphon with a heavily feature-engineered architecture. Therefore, the rebuttal did not significantly change my evaluation.

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Rebuttal by Authors

Rebuttal
by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))

31 Mar 2026, 15:36 (modified: 31 Mar 2026, 22:24)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=KCQSHC78mU)

Rebuttal:
We thank the reviewer for helpful comments.

1. The constraint is handled through a scalar shift in the last layer, and it is unclear how this would generalize to constraints other than edge density.

See our response to Q4 of Reviewer yUB7.

2. The framing of the architecture as being “diffusion-inspired” (...) feels overstated, the architecture is more accurately described as a hand-crafted multiscale sinusoidal neural network.

In the revised version, we will replace "diffusion-inspired" with a more accurate description such as "multiscale sinusoidal residual architecture." We will also motivate the design more directly: implicit neural networks that map graphon coordinates (x, y) to function values tend to learn coarse, low-frequency structure more easily than sharp, high-frequency structure, while many extremal graphons are close to step functions with sharp interfaces. The multiscale sinusoidal components are therefore an intentional domain-specific inductive bias for this class of problems, not evidence of a close formal connection to diffusion models or of general superiority as an implicit representation.

This domain specificity is less limiting than it appears, primarily because extremal graph theory is inherently a study of hard problems with very few constraints. Foundational examples such as Turán's theorem and Sidorenko's conjecture can be reformulated as graphon optimisations with a single constraint, while Ramsey multiplicity problems remain completely unconstrained. Similar variational problems arise in the large-deviation analysis of dense random graphs due to Chatterjee and Varadhan. Thus, our method addresses a broad mathematical class rather than curated examples, and our experiments are intended as representative case studies within that broader class. Our goal is to provide a flexible exploratory optimiser that can generate plausible candidates and hypotheses for the open instances of these problems.

3. The paper does not make a convincing case that a neural representation is preferable to simpler parameterizations such as SBMs, wavelets, or maybe even heuristics like decision trees (...).

See our response to Q3 of Reviewer Baz9.

4. The exposition reads as somewhat unfocused. Lemma 3.1 appears overemphasized compared to its actual role (the work does not seem to use this explicit formula anywhere else, and an autodiff framework can calculate this as well).

Lemma 3.1 justifies the custom backward pass *used* in our constraint solver via the implicit function theorem. If one instead differentiates a naive implementation of Newton's method in JAX or PyTorch, autodiff will by default backpropagate through the unrolled iterations and store their intermediate states, which can be substantially more expensive in memory and time. In that sense, our use of Lemma 3.1 is similar in spirit to implicit-differentiation methods used for implicitly defined models such as deep equilibrium models and neural ODEs.

5. Could you maybe comment more explicitly on where you see this method relative to existing tools, and in what sense you might expect this to assist progress in mathematics?

Flagmatic and related flag-algebra tools are complementary to our approach. For a graphon-minimisation problem, they provide rigorous *lower* bounds together with certificates, whereas our method searches directly for candidate graphons and provides candidate *upper* bounds and candidate minimisers. When an upper bound from our method matches, or nearly matches, a rigorous lower bound, this gives strong evidence for the correct optimum and, importantly, provides a concrete graphon whose structure can be analysed. In addition, SDP-based flag-algebra computations become expensive as the motif graph size grows; for example, Flagmatic 1.5 supports 2-graphs only up to 8 admissible vertices, whereas our method can still be run on larger motif graphs as in the Heawood-vs.-Desargues experiment.

Tools based on combinatorial search such as tabu search are closer in spirit to our goal of discovering good constructions, but they typically operate on finite graphs of fixed size. By contrast, our method optimises directly over graphons and targets asymptotic extremal questions from the start. A practical advantage is interpretability: the learnt graphons can be visualised as 2-D functions, and their coarse structure can often be read off and turned into mathematical conjectures.

We view our method as useful in two main ways. First, it can serve as an exploratory tool that suggests candidate minimisers and structural conjectures for graphon-optimisation problems. Second, it can be used together with rigorous lower-bound methods: if our candidate upper bound nearly matches a certified lower bound, that sharply narrows the search space and can guide the subsequent proof. Our goal is not to replace symbolic methods, but to complement them by generating plausible candidates and hypotheses for open problems.

Replying to Rebuttal by Authors

Rebuttal Acknowledgement by Reviewer uEHG

Rebuttal Acknowledgement by Reviewer uEHG 03 Apr 2026, 23:34

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (b) Partially resolved - I have follow-up questions for the authors.

Reasons:

I thank the authors for their rebuttal, which helped clarify the intended contribution of the paper and the relationship to existing tools. In my view, the architecture being very feature-engineered for this specific problem would not be a concern if there were any signs this approach could help advance extremal graph theory in any meaningful way. However, beyond recovering simple known examples in proof-of-concept experiments, the main experimental claim still rests on table 8 / fig. 8. As I noted in my review (this was also raised by reviewer yUB7), the evidence that this constitutes an actual counterexample is *very* weak. The authors' rebuttal to reviewer yUB7 indicates that the authors will tone down the current wording and refer to this instead as “evidence for a candidate counterexample”. In my opinion, even that characterization is too strong. I do not see how any statistically valid conclusions could be drawn from table 8.

Replying to Rebuttal Acknowledgement by Reviewer uEHG

Reply Rebuttal Comment by Authors

Reply Rebuttal Comment

by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))

07 Apr 2026, 21:26 (modified: 07 Apr 2026, 22:05)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=uxg5phN0rJ)

Comment:

We remind the reviewer that we made the following new experimental findings in extremal graph theory using our approach, which are described in pages 7,8 of the paper.

- Tab 3, Fig 3: On the problem of minimising the C_5 density under a fixed edge density p, our approach found candidate optimal graphons with unexpected banded structure for $p \neq 1 - 1/k$. See Fig 3(d)-(f). Understanding the structure of an optimal graphon is an important topic in extremal graph theory. For the $p = 1 - 1/k$ case, the approach found experimental support for the conjectured optimal C_5 density by Bennett et al. (2020) (https://arxiv.org/abs/1803.00165 (https://arxiv.org/abs/1803.00165)), and provided experimental evidence that an optimal graphon in that case is likely to have the k-partite structure. See Fig 3(a)-(c).
- Tab 4, Fig 4: Using our approach, we analysed the C_7-density minimisation problem under a fixed edge-density constraint for the first time to our knowledge. We generalised the conjecture of Bennett et al. on the C_5 density (described in Appendix D). Using our approach, we found experimental support for the conjecture. Also, candidate optimal graphons for edge densities $p = 1 - 1/k$ found by our approach have the k-partite structure (Fig 4(a)-(c)), but those for $p \neq 1 - 1/k$ have non-trivial structure, such as banded structure (Fig 4(d)-(f)).
- Tab 5, Fig 5: Our study of the curve $p \longmapsto \min_{W:t(K_2,W)=p} t(H, W)$ for a range of graphs H provides a new experimental observation about the shape of this curve; when H is close to being a complete graph, the curve tends to become concave and this concavity fails as H gets further from being complete. This suggests an interesting new question about mathematically characterising the graph H that makes the curve concave. Also, our experiments for $H = H_6$ in Fig. 5(a) show that optimal graphons for H_6 and $p = 1 - 1/k$ are not achieved by k-partite graphons or constant graphons.
- Tab 9, Fig 9, Appendix E: As far as we know, the minimisation of the Petersen density under an edge-density constraint has not been studied before. Our approach experimentally provides information about what optimal graphons might look like and where the optimal values lie.

- Large Deviation Application: During the rebuttal period, we applied our approach to the following graphon-optimisation problem, which is studied by (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) in order to analyse conditioned random graph models:

  $$\inf_W \int \left(W(x,y) \log \frac{W(x,y)}{p} + (1 - W(x,y)) \frac{1 - W(x,y)}{1-p} \right) dx\, dy \,\text{s.t.}\, t(K_3, W) \geq r^3.$$

  By Chatterjee and Varadhan's large deviation theory, the optimising graphons characterise the probability distribution of the Erdős–Rényi random graph $G(n,p)$ conditioned on the triangle density being at least r^3. We applied our approach to $(p,r) \in \{0.05, 0.1\} \times \{0.4, 0.5, 0.6\}$, and consistently found a two-block structured graphon (also known as stochastic block model) as a candidate optimal solution, and such a two-block graphon outperformed the example in (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) experimentally. Our finding gives rise to a conjecture that a two-block graphon is an optimiser. Details of our experiments are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba (https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba)

Regarding H.-D., we will tone down our claim, avoiding the words "evidence" or "counterexample", and will simply say that the found graphon is a candidate optimal solution of $\min_W t(H,W) - t(D,W)^{v(H)/v(D)}$. We think that understanding the structure of an optimising graphon here is valuable because of the importance of the Desargues graph in this context. The Desargues graph satisfies a weakly norming property, i.e., the homomorphism density function $t(H, \cdots)$ is convex on the set of graphons (https://arxiv.org/abs/1611.05784 (https://arxiv.org/abs/1611.05784)). Weakly-norming graphs are known to dominate all of their subgraphs F in the sense that $t(H,W)^{1/e(H)} \geq t(F,W)^{1/e(F)}$. So, weakly norming graphs are positioned close to the top of domination relations and are strong candidates for disproving the conjectured domination inequality. Moreover, if the conjectured domination inequality holds for the Desargues graph, it also implies that the Heawood graph is Sidorenko, $t(H,W) \geq t(K_2,W)^{e(H)}$, concluding one of the hardest instances in the long-lasting conjecture in extremal graph theory. A 3-regular weakly norming graph with girth at least 6 is extremely rare; perhaps very few exist and the Desargues graph seems to be the only one that is small enough to be handled.

Official Review of Submission10609 by Reviewer yUB7

Official Review by Reviewer yUB7 10 Mar 2026, 21:51 (modified: 08 Apr 2026, 21:05)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yUB7

Revisions (/revisions?id=5qIWCGwvkA)

Summary:
The paper proposes using a neural network to represent and optimize graphons in order to automatically discover optimal or near-optimal solutions to open problems in extremal graph theory, using gradient descent with a novel constraint enforcement mechanism and a discontinuity-aware architecture.

Strengths And Weaknesses:
Strengths:

1. Embedding Newton's method in the forward pass with implicit differentiation is mathematically elegant and provably correct.
2. The multi-scale sinusoidal encoding is directly motivated by the mathematical structure of optimal graphons being step functions.
3. Within 0.1% of proven optima across Tables 1-4 for small graphs. For a gradient descent method with no theoretical convergence guarantees this is remarkably good.

Weaknesses:

1. One model per (H, p) pair trained from scratch. No transfer learning across problems. For 875 instances in the exhaustive study this is extremely expensive and doesn't scale.
2. Table 8 exposes the fundamental scaling problem. For Heawood and Desargues graphs the Monte Carlo estimates are at 10^{-25} scale with confidence intervals spanning 6x. The method essentially fails to give

meaningful results for large H.

3. 11.8% of instances in the exhaustive study clearly failed or were suboptimal. No principled way to know if more computation would help or if the problem is fundamentally harder.
4. This work only handles $t(K_2, W) = p$. Real extremal problems often involve multiple simultaneous constraints. Extending to multiple constraints requires solving a system rather than a scalar Newton step, which is significantly harder.

Soundness: 3: good
Presentation: 3: good
Significance: 3: good
Originality: 3: good
Key Questions For Authors:
None

Limitations:
Yes

Overall Recommendation: 4: Weak accept: Technically solid paper that advances at least one sub-area of AI, with a contribution that others are likely to build on, but with some weaknesses that limit its impact (e.g., limited evaluation). Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Compliance With LLM Reviewing Policy: Affirmed.
Code Of Conduct Acknowledgement: Affirmed.
Final Justification:
Despite my remaining reservations regarding the statistical significance of the results in Table 8, I am moving to a Weak Accept. While the authors' response on this specific point was not entirely convincing, I believe the overall architectural contributions and the insights provided in other parts of the rebuttal merit giving the paper a chance. I trust the authors will make a good-faith effort to improve the empirical reporting in the final version.

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Rebuttal by Authors

Rebuttal
by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))
31 Mar 2026, 15:37 (modified: 31 Mar 2026, 22:24)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors
Revisions (/revisions?id=sm9dASvgLR)

Rebuttal:
We thank the reviewer for helpful comments.

1. One model per (H, p) pair trained from scratch. No transfer learning across problems. For 875 instances in the exhaustive study this is extremely expensive and doesn't scale.

The exhaustive study is indeed computationally intensive, but the 875 (H, p) instances are fully parallelisable. In our setup, we completed the full sweep in under two weeks on five RTX A6000 GPUs, which are not as high-performing as A100 or H100. We agree, however, that the current method is instance-wise and does not yet exploit shared structure across related tasks; we will make this limitation clearer in the revision.

We agree that multi-task learning is a promising direction. To probe this point, during the rebuttal period we ran a preliminary FiLM-conditioned experiment that efficiently and jointly trains across multiple edge densities $p \in [\frac{1}{2}, \frac{2}{3}]$ for the fixed $H = K_3$. The learnt graphons achieve absolute error at most 2.09×10^{-4} and average absolute error 3.12×10^{-5} relative to the known optimum, with p sampled uniformly from that interval. This does not yet extend across different graphs, so we view it as preliminary evidence that shared training across related tasks can reduce the per-instance burden. We will add this experiment to the revision and clearly label it as preliminary. Per-p errors are available at the following anonymous link: https://acrobat.adobe.com/id/urn:aaid:sc:AP:b09510ec-6b09-44b6-8ebe-bbb73c9bb93a (https://acrobat.adobe.com/id/urn:aaid:sc:AP:b09510ec-6b09-44b6-8ebe-bbb73c9bb93a)

2. Table 8 exposes the fundamental scaling problem. For Heawood and Desargues graphs the Monte Carlo estimates are at 10^{-25} scale with confidence intervals spanning 6x. The method essentially fails to give meaningful results for large H.

Although the absolute scale of the quantities in Table 8 is around 10^{-25}, the difference is of the same order as the densities $t(H, W)$ and $t(D, W)^{v(H)/v(D)}$ themselves, so it is not negligible in relative terms. More importantly, floating-point representation is not the bottleneck here: double precision (float64) easily represents numbers at this scale, and the dominant uncertainty is statistical rather than numerical.

As indicated by the confidence intervals and the non-negativity of the conservative difference in Table 8, our current estimates of $t(H, W) - t(D, W)^{v(H)/v(D)}$ are not yet precise enough to definitively conclude that the inequality $t(H, W) \geq t(D, W)^{v(H)/v(D)}$ is false. We will make this limitation clearer in the revised version and present this result as evidence for a candidate counterexample rather than a definitive disproof.

3. 11.8% of instances in the exhaustive study clearly failed or were suboptimal. No principled way to know if more computation would help or if the problem is fundamentally harder.

We should clarify that "failure/suboptimality" is an operational label rather than a formal certificate of fundamental hardness. In the exhaustive study, we used this label when, for a fixed graph H, the candidate upper bound at a particular edge density p failed to match the lower bound computed by our flag-algebra implementation and noticeably broke the empirical trend observed across nearby values of p for the same H; for example, monotonicity or approximate convexity after interpolation. For every H tried, such anomalous cases occurred at most two times. We conjecture that these cases are primarily due to optimisation failures and worth revisiting with more computation, rather than as evidence that the underlying problem is intrinsically harder. We will revise the paper to make this heuristic criterion explicit.

4. This work only handles $t(K_2, W) = p$. Real extremal problems often involve multiple simultaneous constraints.

For a single constraint, the solver is more general than the edge-density case: it applies whenever the constraint can be enforced by solving for one scalar control parameter, for example constraints of the form $t(H, W) = p$ or, more generally, $f(t(H, W)) = r$ for a monotone function f and a graph H (which may be different from K_2). We do not claim that the current solver handles multiple constraints exactly. A practical workaround is to enforce one constraint through the solver and treat the others as penalties, but a principled extension would require a vector-valued root-finding procedure or a projection layer, which we view as important future work.

At the same time, we note that many canonical graphon-optimisation problems in extremal graph theory fall into the one-constraint or unconstrained setting, including the Turán-type, Sidorenko-type, and Ramsey multiplicity problems. We will clarify in the revision that the current paper is intentionally focused on this single-constraint regime rather than claiming a complete treatment of general multi-constraint extremal optimisation.

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Replying to Rebuttal by Authors

Rebuttal Acknowledgement by Reviewer yUB7

Rebuttal Acknowledgement by Reviewer yUB7 03 Apr 2026, 01:27

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (b) Partially resolved - I have follow-up questions for the authors.

Reasons:

For the second weakness, I get that float64 handles the scale, but that's not the issue. The real problem is the statistical noise. If your confidence interval is 6x larger than the estimate itself, the "signal" is essentially buried. In Monte Carlo terms, a margin of error that wide means we can't even be sure of the sign of the result—it could just as easily be zero or negative.

Replying to Rebuttal Acknowledgement by Reviewer yUB7

Reply Rebuttal Comment by Authors

Reply Rebuttal Comment

by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))

07 Apr 2026, 21:25 (modified: 07 Apr 2026, 22:05)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=LL3Et8Fzbp)

Comment:

Regarding H.-D., we will tone down our claim, avoiding the words "evidence" or "counterexample", and will simply say that the found graphon is a candidate optimal solution of $\min_W t(H,W) - t(D,W)^{v(H)/v(D)}$. We think that understanding the structure of an optimising graphon here is valuable because of the importance of the Desargues graph in this context. The Desargues graph satisfies a weakly norming property, i.e., the homomorphism density function $t(H,\cdots)$ is convex on the set of graphons (https://arxiv.org/abs/1611.05784 (https://arxiv.org/abs/1611.05784)). Weakly-norming graphs are known to dominate all of their subgraphs F in the sense that $t(H,W)^{1/e(H)} \geq t(F,W)^{1/e(F)}$. So, weakly norming graphs are positioned close to the top of domination relations and are strong candidates for disproving the conjectured domination inequality. Moreover, if the conjectured domination inequality holds for the Desargues graph, it also implies that the Heawood graph is Sidorenko, $t(H,W) \geq t(K_2,W)^{e(H)}$, concluding one of the hardest instances in the long-lasting conjecture in extremal graph theory. A 3-regular weakly norming graph with girth at least 6 is extremely rare; perhaps very few exist and the Desargues graph seems to be the only one that is small enough to be handled.

Before finishing, we re-emphasise that we made the following new experimental findings in extremal graph theory using our approach described in pages 7 and 8 of the paper.

- Tab 3, Fig 3: On the problem of minimising the C_5 density under a fixed edge density p, our approach found candidate optimal graphons with unexpected banded structure for $p \neq 1 - 1/k$. See Fig 3(d)-(f). Understanding the structure of an optimal graphon is an important topic in extremal graph theory. For the $p = 1 - 1/k$ case, the approach found experimental support for the conjectured optimal C_5 density by Bennett et al. (2020) (https://arxiv.org/abs/1803.00165 (https://arxiv.org/abs/1803.00165)), and provided experimental evidence that an optimal graphon in that case is likely to have the k-partite structure. See Fig 3(a)-(c).
- Tab 4, Fig 4: Using our approach, we analysed the C_7-density minimisation problem under a fixed edge-density constraint for the first time to our knowledge. We generalised the conjecture of Bennett et al. on the C_5 density (described in Appendix D). Using our approach, we found experimental support for the conjecture. Also, candidate optimal graphons for edge densities $p = 1 - 1/k$ found by our approach have the k-partite structure (Fig 4(a)-(c)), but those for $p \neq 1 - 1/k$ have non-trivial structure, such as banded structure (Fig 4(d)-(f)).
- Tab 5, Fig 5: Our study of the curve $p \longmapsto \min_{W:t(K_2,W)=p} t(H,W)$ for a range of graphs H provides a new experimental observation about the shape of this curve; when H is close to being a complete graph, the curve tends to become concave and this concavity fails as H gets further from being complete. This suggests an interesting new question about mathematically characterising the graph H that makes the curve concave. Also, our experiments for $H = H_6$ in Fig. 5(a) show that optimal graphons for H_6 and $p = 1 - 1/k$ are not achieved by k-partite graphons or constant graphons.
- Tab 9, Fig 9, Appendix E: As far as we know, the minimisation of the Petersen density under an edge-density constraint has not been studied before. Our approach experimentally provides information about what optimal graphons might look like and where the optimal values lie.

- Large Deviation Application: During the rebuttal period, we applied our approach to the following graphon-optimisation problem, which is studied by (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) in order to analyse conditioned random graph models:

$$\inf_W \int \left(W(x,y) \log \frac{W(x,y)}{p} + (1-W(x,y)) \frac{1-W(x,y)}{1-p} \right) dx\, dy \,\text{s.t.}\; t(K_3, W) \geq r^3.$$

By Chatterjee and Varadhan's large deviation theory, the optimising graphons characterise the probability distribution of the Erdős–Rényi random graph $G(n,p)$ conditioned on the triangle density being at least r^3. We applied our approach to $(p,r) \in \{0.05, 0.1\} \times \{0.4, 0.5, 0.6\}$, and consistently found a two-block structured graphon (also known as stochastic block model) as a candidate optimal solution, and such a two-block graphon outperformed the example in (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) experimentally. Our finding gives rise to a conjecture that a two-block graphon is an optimiser. Details of our experiments are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba (https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba)

Official Review of Submission10609 by Reviewer Baz9

Official Review by Reviewer Baz9 08 Mar 2026, 18:47 (modified: 07 Apr 2026, 12:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Baz9

Revisions (/revisions?id=qcdIK0ngy9)

Summary:

The paper proposes a neural approach to asymptotic extremal graph theory by optimizing over graphons represented as implicit neural networks. The key ideas are a diffusion-inspired, multi-scale sinusoidal residual architecture tailored to capture discontinuous, step-like optimal graphons, an embedded empirical constraint solver (for fixed edge density) differentiated via implicit differentiation, and a variance-reduced estimator of homomorphism densities that exploits graph symmetries. Empirically, the method rediscovers several known optima (e.g., triangle and clique densities at specific edge densities), closely matches conjectured minima for C5 and proposes candidates for C7, and yields potential counterexamples to certain normalised homomorphism-density inequalities; it also conducts an exhaustive sweep on small graphs and presents ablations supporting the architectural choices.

Strengths And Weaknesses:

Strength The integration of a per-layer multi-scale sinusoidal input encoding with residual connections is a well-motivated inductive bias for step-like graphons, distinct from standard MLP implicit representations. The embedded constraint solver for edge density with implicit differentiation is an elegant mechanism to satisfy constraints during training without fragile penalty tuning.

Weakness The constraint-solver mechanism is specialized to a single scalar constraint (edge density) by adjusting a global bias; extending to multiple/complex constraints is not demonstrated. The empirical constraint is only satisfied in expectation over batches; the learned W may deviate from exact t(K2,W)=p at the graphon level without a projection step. Missing comparisons to strong non-neural baselines tailored to step graphons (e.g., variable-k stochastic block model or piecewise-constant parametric families optimised by gradient/EM), which would be natural and potentially competitive on these problems.

Soundness: 3: good
Presentation: 3: good
Significance: 2: fair
Originality: 3: good
Key Questions For Authors:
7.Rebuttal questions How general is the constraint-solver approach beyond the single edge-density constraint? Can your framework handle multiple constraints (e.g., fixing t(F1,W), t(F2,W) simultaneously) without changing the architecture substantially? For the inequality counterexamples (K_r vs K_s and Heawood vs Desargues), how robust are the negative gaps under: (i) increased sample sizes, (ii) different random seeds/initializations, and (iii) finer discretization grids or purely Monte Carlo estimates with tight CIs?

Limitations:

Yes

Overall Recommendation: 4: Weak accept: Technically solid paper that advances at least one sub-area of AI, with a contribution that others are likely to build on, but with some weaknesses that limit its impact (e.g., limited evaluation). Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Compliance With LLM Reviewing Policy: Affirmed.
Code Of Conduct Acknowledgement: Affirmed.

Rebuttal by Authors

Rebuttal
by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))
31 Mar 2026, 15:37 (modified: 31 Mar 2026, 22:24)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors
Revisions (/revisions?id=8BhEMOHepK)

Rebuttal:

1. The constraint-solver mechanism is specialized to a single scalar constraint (edge density) (...); extending to multiple/complex constraints is not demonstrated.

For a single constraint, the solver is more general than the edge-density case: it applies whenever the constraint has the form $f(t(H,W)) = r$ for a monotone function f, a real r, a general graph H. We do not claim that our solver handles multiple constraints. A practical workaround is to enforce one constraint through the solver and treat the others as penalties, but a principled extension would require a vector-valued root-finding procedure or a projection layer, which we view as important future work.

We note that many classical graphon-optimisation problems in extremal graph theory fall into the one-constraint or unconstrained setting, including Turan-type, Sidorenko-type, and Ramsey multiplicity problems. We will clarify in the revision that the paper is intentionally focused on this single-constraint regime rather than claiming a complete treatment of general multi-constraint extremal optimisation.

2. The empirical constraint is only satisfied in expectation over batches (...).

We will state this caveat more precisely in the revised version. During training, our solver enforces the constraint for the mini-batch estimator rather than performing an exact projection onto $W : t(K_2, W) = p$ at the graphon level. At evaluation time, we re-solve the constraint equation for the bias term c, namely the equation that makes the empirical estimate of $t(K_2, W)$ equal to the target p, but using a much larger sample budget. We then report edge densities after large-sample or discretised reevaluation, so any residual mismatch is reduced to numerical estimation error. Of course, we do not claim exact graphon-level satisfaction. We will revise the wording to make that explicit.

3. Missing comparisons to strong non-neural baselines tailored to step graphons (...), which would be natural and potentially competitive on these problems.

During the rebuttal, we evaluated non-neural baselines on the task used in our ablation study: minimising triangle density under the edge-density constraint $p = 7/9$. The baselines were: (a) an SBM with 256 equal-sized blocks, with trainable edge densities; (b) SBMs with 5 blocks and 64 blocks, with trainable edge densities and block sizes; (c) a decision tree; (d) a tree ensemble. Each baseline was run four times with different initialisations, and we report the best objective value across runs. The SBMs with trainable block sizes collapsed to nearly constant graphons, suggesting a difficult optimisation landscape. The fixed-size SBM recovered a noisy multi-block structure. The tree-based baselines produced structured graphons, but they remained qualitatively different from the solutions found by our method and achieved worse objective values. Overall, all of these baselines performed worse than even the ablated variants of our method on this task. Detailed results are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:faea5c27-a686-4210-82b4-1c5af8d5de19 (https://acrobat.adobe.com/id/urn:aaid:sc:AP:faea5c27-a686-4210-82b4-1c5af8d5de19)

4. How general is the constraint-solver approach beyond the single edge-density constraint? Can your framework handle multiple constraints (...) without changing the architecture substantially?

As explained in our answer to Q1, our solver is more general than the edge-density case: it can handle single constraints of the form $f(t(H, W)) = r$ for a monotone function f, a graph H, a real r. Extending it to multiple simultaneous constraints is substantially harder and would require modifying the solver, for example through a vector-valued root-finding procedure or a projection-based method. We view multiple-constraint handling as important future work rather than part of the current contribution.

5. For the inequality counterexamples (...), how robust are the negative gaps under (...)?

Our robustness checks suggest that different random initialisations of the network parameters have relatively little qualitative effect, whereas batch size matters much more. In particular, smaller batch sizes (256 to 1024) made it harder for the trained networks to learn sharp boundaries, which in turn weakened the observed negative gaps. During the rebuttal, we reevaluated the graphons found by our approach for the K_r vs. K_s inequalities while varying the number N of Monte Carlo samples over $\{2^k\}_{k=10}^{28}$ and the discretisation resolution R over $\{2^d\}_{d=3}^{10}$. The results are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:e16965c4-c0f3-4a78-b77b-55c562378c06 (https://acrobat.adobe.com/id/urn:aaid:sc:AP:e16965c4-c0f3-4a78-b77b-55c562378c06). For discretisation, high resolution was not needed: $R = 2^4$ already yielded a clear negative gap, and $R = 2^8$ was sufficient to estimate the target value accurately. For Monte Carlo, by contrast, obtaining a stable estimate with a reasonably tight confidence interval required at least 2^{24} samples. We will add this discussion to the revised version.

Replying to Rebuttal by Authors

Rebuttal Acknowledgement by Reviewer Baz9

Rebuttal Acknowledgement by Reviewer Baz9 01 Apr 2026, 14:37

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (b) Partially resolved - I have follow-up questions for the authors.

Reasons:

In response to the author's rebuttal, I maintain my original score.

Replying to Rebuttal Acknowledgement by Reviewer Baz9

Reply Rebuttal Comment by Authors

Reply Rebuttal Comment

by Authors (Hongseok Yang (/profile?id=~Hongseok_Yang2), Taeyoung Kim (/profile?id=~Taeyoung_Kim2), Jineon Baek (/profile?id=~Jineon_Baek1), Joonkyung Lee (/profile?id=~Joonkyung_Lee1))

07 Apr 2026, 21:26

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

We re-emphasise that we made the following new experimental findings in extremal graph theory using our approach described in pages 7 and 8 of the paper.

- Tab 3, Fig 3: On the problem of minimising the C_5 density under a fixed edge density p, our approach found candidate optimal graphons with unexpected banded structure for $p \neq 1 - 1/k$. See Fig 3(d)-(f). Understanding the structure of an optimal graphon is an important topic in extremal graph theory. For the $p = 1 - 1/k$ case, the approach found experimental support for the conjectured optimal C_5 density by Bennett et al. (2020) (https://arxiv.org/abs/1803.00165 (https://arxiv.org/abs/1803.00165)), and provided experimental evidence that an optimal graphon in that case is likely to have the k-partite structure. See Fig 3(a)-(c).
- Tab 4, Fig 4: Using our approach, we analysed the C_7-density minimisation problem under a fixed edge-density constraint for the first time to our knowledge. We generalised the conjecture of Bennett et al. on the C_5 density (described in Appendix D). Using our approach, we found

experimental support for the conjecture. Also, candidate optimal graphons for edge densities $p = 1 - 1/k$ found by our approach have the k-partite structure (Fig 4(a)-(c)), but those for $p \neq 1 - 1/k$ have non-trivial structure, such as banded structure (Fig 4(d)-(f)).
- Tab 5, Fig 5: Our study of the curve $p \longmapsto \min_{W:t(K_2,W)=p} t(H, W)$ for a range of graphs H provides a new experimental observation about the shape of this curve; when H is close to being a complete graph, the curve tends to become concave and this concavity fails as H gets further from being complete. This suggests an interesting new question about mathematically characterising the graph H that makes the curve concave. Also, our experiments for $H = H_6$ in Fig. 5(a) show that optimal graphons for H_6 and $p = 1 - 1/k$ are not achieved by k-partite graphons or constant graphons.
- Tab 9, Fig 9, Appendix E: As far as we know, the minimisation of the Petersen density under an edge-density constraint has not been studied before. Our approach experimentally provides information about what optimal graphons might look like and where the optimal values lie.
- Large Deviation Application: During the rebuttal period, we applied our approach to the following graphon-optimisation problem, which is studied by (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) in order to analyse conditioned random graph models:

$$\inf_W \int \left(W(x,y) \log \frac{W(x,y)}{p} + (1 - W(x,y)) \frac{1 - W(x,y)}{1 - p} \right) dx\, dy \text{ s.t. } t(K_3, W) \geq r^3.$$

By Chatterjee and Varadhan’s large deviation theory, the optimising graphons characterise the probability distribution of the Erdős–Rényi random graph $G(n, p)$ conditioned on the triangle density being at least r^3. We applied our approach to $(p, r) \in \{0.05, 0.1\} \times \{0.4, 0.5, 0.6\}$, and consistently found a two-block structured graphon (also known as stochastic block model) as a candidate optimal solution, and such a two-block graphon outperformed the example in (https://arxiv.org/abs/1210.7013 (https://arxiv.org/abs/1210.7013)) experimentally. Our finding gives rise to a conjecture that a two-block graphon is an optimiser. Details of our experiments are available at https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba (https://acrobat.adobe.com/id/urn:aaid:sc:AP:b6e12c4c-1ae6-49c0-b3ed-7eef102a90ba)

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HIGH LEVEL SUMMARY

Paper Summary

The paper "Implicit Neural Representation of Graphons for Extremal Graph Theory" introduces a deep learning framework to address asymptotic problems in extremal graph theory. These problems are reformulated as continuous optimization tasks over graphons, which the authors represent using Implicit Neural Representations (INRs). To address the challenge that optimal graphons are often discontinuous (step-like), the proposed architecture utilizes a diffusion-inspired design with multi-scale sinusoidal input encodings. For constrained optimization problems (e.g., fixed edge density), the method incorporates a numerical solver into the architecture and employs implicit differentiation for backpropagation. The approach is evaluated by demonstrating its ability to rediscover known optima for generalized Turán problems and by proposing novel candidate optima and counterexamples for open problems, including C_5 and C_7 minimization and inequalities involving the Heawood and Desargues graphs.

Key Issues Roadmap

- **[1. Introduction and Background]**: Potential lack of precision in the formulation of the discrete asymptotic optimization problem (Eq. 4), as the exact edge density constraint may not be feasible for finite graphs for arbitrary p.
- **[2. Methodology]**: Potential issues with the correctness and clarity of the methodology, specifically the initialization of sinusoidal encoding weights (L172-173) which appears deterministic as written, and ambiguous notation (unbound indices) in the density estimator (Eq. 10) and its gradient.
- **[3. Experimental Evaluation]**: Concerns regarding the interpretation of results: the statistical metric used for the Heawood vs. Desargues candidate (Table 8) does not confirm the counterexample with high confidence, and there is a statistically significant gap between achieved results and known optima for K_k minimization (Table 2).
- **[4. Related Work, Proofs, and Appendices]**: Potential critical mathematical errors in the appendices. The edge density constraint formulation in Appendix D (L735) appears dimensionally incorrect (containing a cubic term), potentially impacting the reported conjectured optima. Additionally, the definitions of baseline graphons W3 and W4 in Appendix F appear invalid (resulting in densities > 1) for the specified parameter regime.

DETAILED SEGMENT REVIEWS

[1] SEGMENT: 1. Introduction and Background

PAGES: [[1, 2]]

1. Summary

The segment reviewed covers the Introduction (Section 1) and Background (Section 2). Section 1 motivates the work by situating it within the context of using machine learning for mathematical discovery, specifically targeting asymptotic problems in extremal graph theory. It introduces the concept of reformulating these problems as continuous optimization over graphons and proposes the use of implicit neural representations (INRs) to model these graphons. The authors identify key challenges (discontinuous optima and constraints) and summarize their proposed techniques.

Section 2 provides the necessary mathematical definitions, including graph notation, homomorphism density for finite graphs (Eq. 2), and the general structure of the asymptotic optimization problems (Eq. 3). It highlights two specific problem classes: generalized Turán problems (Eq. 4) and density inequalities related to Sidorenko's conjecture (Eq. 5). The section concludes by defining graphons and extending homomorphism density to this continuous setting (Eq. 6).

2. Potential Mistakes and Improvements

1. **Mathematical Rigor of Eq. (4) Formulation:** The formulation of the asymptotic optimization problem in Eq. (4) lacks precision: $\liminf_{n\to\infty}(\min_G t(H,G) \text{ s.t. } v(G) = n \wedge t(K_2, G) = p)$. The inner minimization requires the edge density $t(K_2, G)$ of a finite graph G with n vertices to exactly equal p. However, edge densities of finite graphs can only take specific discrete rational values. If p is irrational, or a rational number not representable by a graph of size n, the constraint cannot be satisfied. The feasible set for the minimization would be empty, resulting in a minimum of $+\infty$. While the paper subsequently relies on the equivalent graphon formulation (where the constraint is feasible), the formulation of the discrete asymptotic problem in Eq. (4) is technically problematic as written. It may be clearer to formulate this by considering sequences of graphs whose edge densities converge to p, or emphasize that this notation refers to the equivalent graphon optimization problem.
2. **Clarity of Definitions in Eq. (5):** In Eq. (5) (L085), the exponents m_1 and m_2 are used before they are defined. They are later defined as positive integers in Section 3 (L124). To improve clarity, these parameters should be defined upon their introduction in Section 2.
3. **Precision Regarding Sidorenko's Conjecture (L087):** Line 087 states that "Sidorenko's conjecture [...] predicts a positive answer" regarding the question of whether the asymptotic optimum in Eq. (5) is non-negative. The conjecture specifically predicts that the optimum is *non-negative* (≥ 0), not strictly positive (> 0).

3. Minor Corrections and Typos

- L033, L074: "Turan" should be "Turán".
- L041, L077 (and elsewhere): "Lovasz" should be "Lovász".
- L074: "Erdos" should be "Erdős".

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[2] SEGMENT: 2. Methodology: Implicit Neural Representation of Graphons

PAGES: [[3, 5]]

1. Summary Section 3 (Pages 3-5) details the methodology for optimizing graphons represented as Implicit Neural Networks (INRs), targeting the optimization problems formalized in Eq. (8). The authors introduce an architecture designed to handle the discontinuous (step-like) nature of optimal graphons: a residual MLP incorporating multi-scale sinusoidal encodings with increasing frequency scales in deeper layers. To handle constraints (e.g., fixed edge density), the method embeds a numerical solver (Eq. 9) that determines a scalar offset parameter c at each forward pass. Backpropagation through this solver is performed via implicit differentiation (Lemma 3.1). The section also details an estimator for the homomorphism density (Eq. 10), which involves averaging over a set of permutations S for potential variance reduction and efficiency gains.

2. Potential Mistakes and Improvements

- **Correctness: Initialization of Sinusoidal Encoding Weights (P3, L172-173)** The initialization for the input encoding matrices $U^{(\ell)}$ is specified as $U_{i,k}^{(\ell)} \overset{\text{iid}}{\sim} \mathrm{Unif}[s(\ell)/\sqrt{2}, s(\ell)/\sqrt{2}]$. A uniform distribution $\mathrm{Unif}[a,a]$ denotes a deterministic value a. This implies all entries of $U^{(\ell)}$ are initialized to the constant $s(\ell)/\sqrt{2}$. This appears inconsistent with the goal of introducing diverse sinusoidal features and the "iid" notation. Clarification of the intended initialization distribution is necessary for correctness and reproducibility.
- **Correctness/Clarity: Ambiguous Notation in Density Estimator (P4, Eq. 10)** The notation for the homomorphism density estimator in Eq. (10) is mathematically ambiguous: $$ \hat{t}(H, f_{\theta}[c]) := \frac{1}{N|S|} \sum_{k\in[N]} \sum_{\pi\in S} \prod_{{\pi(i),\pi(j)} \in E(H)} f_{\theta}c (x%5E%7B(k)%7D_%7B%5Cpi(i)}, x^{(k)}*{\pi(j)}). $$ The indices i and j under the product operator are unbound, making the expression ill-defined. Assuming the intention is to permute the assignment of samples to the vertices of H, the notation requires correction, perhaps as $\prod*{{i,j}\in E(H)} f_{\theta}c (x%5E%7B(k)%7D_%7B%5Cpi(i)}, x^{(k)}_{\pi(j)})$. This ambiguity also appears in the gradient expression at the bottom of Page 4 (L219).

- **Clarity/Reproducibility: Estimation of Polynomial Expressions (P4, L191-202 and Footnote 2)** There is ambiguity regarding the practical estimation of polynomial expressions of homomorphism densities. The main text (L191-202) describes expanding the polynomial and using the identity in Eq. (11). However, Footnote 2 (L219) states, "In practice, we use U-statistics of monomial." It is unclear which method was implemented. Clarifying the exact form of the estimator used in experiments is necessary for reproducibility.
- **Clarity: Computational Complexity and Scalability** The methodology involves computationally intensive steps at every iteration: Monte Carlo estimation of homomorphism densities (which generally scales poorly with the size of H) and an iterative constraint solver (Eq. 9) with subsequent implicit differentiation. The methodology section would benefit from an analysis of the computational complexity or a discussion of the practical scalability of the approach regarding the size of H and the overhead of the embedded solver.
- **Design Rationale: Justification for Variance Reduction (P4, L181-184)** The paper claims that the estimator (Eq. 10) often achieves "substantially lower variance" than the standard Monte Carlo estimator "at the same sampling cost." This strong claim is presented without theoretical justification or empirical support in the main text. Additionally, "same sampling cost" is ambiguous; it should be clarified if this refers to the number of tuples N or the total computational budget.
- **Clarity: Stability and Implementation of the Constraint Solver (P3, L183-184)** The text mentions using a "variant of Newton's method" to solve Eq. (9). Appendix C reveals that standard Newton's method is unstable and a specific clipped variant is required. While details are provided in the appendix, briefly mentioning the need for stabilization techniques in Section 3 would better characterize the methodology's complexity and implementation requirements.

3. Minor Corrections and Typos

- None identified.

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[3] SEGMENT: 3. Experimental Evaluation and Discoveries

PAGES: [[5, 8]]

1. Summary Section 4 presents the empirical evaluation of the proposed implicit neural representation method for graphon optimization. The evaluation focuses on validating the method by rediscovering known optima and investigating open problems to generate new candidates.

The method is validated on constrained minimization of clique densities (K_k). It recovers the known structures and achieves numerical results close to the theoretical optima for K_3 (Table 1). For K_k (k>3), the structures are qualitatively recovered (Fig. 2), although numerical results show a small gap (Table 2).

For open problems concerning C_5 and C_7 minimization, the method yields results consistent with existing (C_5) and newly proposed (C_7) conjectures (Tables 3-4). A systematic study on 175 small graphs compares results against flag algebra bounds and identifies a novel candidate structure for H_6 (Fig. 5). The method is also used to find candidate counterexamples for normalized density inequalities, including clique density inequalities (Table 6) and the Heawood vs. Desargues conjecture (Table 8).

An ablation study (Table 7) assesses the impact of the multi-scale encoding, the explicit constraint solver, and the learning rate schedule.

2. Potential Mistakes and Improvements

1. **Statistical Interpretation of the Heawood vs. Desargues Candidate (P8, Table 8):** The authors investigate a conjecture by searching for a counterexample where A < B (normalized densities of Heawood and Desargues graphs). In Table 8, footnote 'a' defines "Conservative Diff" as (Upper CI_A − Lower CI_B) and states that "A negative value confirms A < B with high confidence." However, the reported Conservative Diff is positive (1.86×10^{-25}). According to the authors' stated metric, this result does not confirm the counterexample with high confidence, as the confidence intervals overlap. The interpretation of this result should be clarified to reflect the statistical uncertainty.
2. **Statistically Significant Suboptimality in K_k Minimization (P5, Table 2):** Table 2 reports results for minimizing K_k density (k=4, 5, 6) where the optimum is known. The reported 95% confidence intervals for the estimated densities are strictly above the known optima. For instance, for k=4, the optimum is 0.09375, while the CI is (0.09403727, 0.09411154). This demonstrates a statistically significant gap between the achieved results and the theoretical optima. This quantitative discrepancy should be acknowledged.
3. **Reproducibility and Fairness of Ablation Study (P8, L430-433):** The ablation study mentions performing hyperparameter searches for the baseline variants (e.g., regularizer strength, constant learning rate) to

ensure a fair comparison. However, the details of these searches (the search spaces explored and the best parameters found) are not reported in the paper or appendix. This information is necessary to assess the fairness of the comparison and ensure reproducibility.

4. **Missing Details for Reproducibility (Architecture and Evaluation):**
    - The architecture definition (L151) allows for two input-encoding scale functions: $s(\ell) = \ell$ or $s(\ell) = 2^{\ell-1}$. It is not specified in Section 4 or Appendix C which function was used for the experiments.
    - The evaluation methodology (L265-269) mentions discretization (M x M matrix) or estimation. For Tables 1, 3-5, and 7, which do not show CIs, the discretization resolution M should be specified to clarify how the reported values were obtained. (M=2^{14} is mentioned only for C_5 at L366).
5. **Ambiguity in Exhaustive Study Comparison (P8, L402-404):** The exhaustive study reports that the method produced a graphon whose H-density "closely matches" the flag-algebra lower bound in 76.5% of instances. Providing a quantitative definition or threshold (e.g., a relative error margin) for "closely matches" would improve the rigor of this claim.
6. **Optimization Stability (P5, L269-270):** The paper reports only the run with the lowest objective value across multiple training runs. To characterize the stability and robustness of the optimization procedure, it is recommended to report the variance (e.g., mean and standard deviation) of the objective values across these runs.

3. Minor Corrections and Typos

- P5, L268: Missing space in `−btD′ (H`.
- P7 (L333, L346) and P8 (L436, L439), Table 6, Figure 6 caption: The notation for binomial coefficients, e.g., (s 2), should be rendered using standard mathematical notation, e.g., $\binom{s}{2}$.
- P8, Table 8: The exponents $v(H)/v(D)$ are poorly rendered and difficult to read.

[4] SEGMENT: 4. Related Work, Proofs, and Appendices

PAGES: [[9, 16]]

1. Summary The reviewed segment includes the Impact Statement, References, and Appendices A through F (Pages 9-16). Appendix A discusses related work, situating the proposed method relative to Flag Algebras and reinforcement learning approaches for discrete optimization. Appendix B provides proofs for the unbiasedness of the homomorphism density estimator (Lemma B.1) and the gradient calculation via implicit differentiation (Proof of Lemma 3.1). These mathematical derivations were verified and appear correct. Appendix C details the experimental setup, hyperparameters, and Algorithm 1, which implements the constraint solver using a clipped Newton's method. Appendix D formalizes the conjecture regarding the optimal graphons for odd-cycle density minimization, providing the associated polynomial forms for C5 and C7 densities. Appendix E presents additional experimental results concerning the Petersen graph. Appendix F details the construction of the hand-crafted graphons used as baselines.

2. Potential Mistakes and Improvements

1. **[Correctness] Appendix D: Incorrect edge density constraint formulation.** On line 735, the edge density constraint for the optimization problem related to the odd-cycle conjecture is stated as: $(k-1)_2x^2 + 2(k-1)xy + \rho y^3 = p$. This equation appears incorrect based on the graphon construction defined in lines 707-712. The edge density $t(K_2, W)$ must be a quadratic function of the partition lengths x and y. The term ρy^3 is cubic and therefore dimensionally inconsistent. Given the construction where Y_1 and Y_2 both have measure $y/2$ and the density between them is ρ, their contribution to the edge density should be $2 \cdot (y/2) \cdot (y/2) \cdot \rho = \rho y^2/2$. The constraint equation requires correction. This error potentially impacts the calculation of the conjectured optima reported in the main paper (e.g., Tables 3 and 4).
2. **[Correctness] Appendix F: Invalid definitions for hand-crafted graphons W3 and W4.** Appendix F defines baseline graphons for edge densities p satisfying $1 - 1/k < p < 1 - 1/(k+1)$ (L805-806). However, the definitions for W3 and W4 result in edge densities greater than 1 in this regime, making them invalid graphons.
    - Graphon W3 (L847-849) assigns an edge density of $kp/(k-1)$ between distinct partitions. For this density to be ≤ 1, it requires $p \leq 1 - 1/k$.
    - Graphon W4 (L850-853) assigns an edge density $C = (pk^2 - k^2 + k + 2)/2$ between X1 and X2. For $C \leq 1$, it also requires $p \leq 1 - 1/k$. Both definitions contradict the premise that $p > 1 - 1/k$. The definition of these baselines, which are used in the experiments (e.g., Table 9), needs to be corrected.

3. **[Clarity] Appendix C, Algorithm 1: Undefined variable in the clipping range.** In Algorithm 1 (L694), the Newton update step uses a clipping range defined as `Clip[−D/t, D/t]`. The variable `t` is not defined or initialized within the pseudocode. Although the accompanying text (L680) mentions that the clipping range is gradually decreased, the algorithm lacks the explicit definition or update rule for `t` (which might be the iteration counter), hindering reproducibility.